

Introduction to Tilings

Let $\mathcal{T} = \{T_\alpha\}_{\alpha \in A}$ be a collection of subsets of the Euclidean plane $\mathbb{R}^2$. We say that $\mathcal{T}$ is a **tiling (or tessellation)** if the following three conditions hold:

1. **Covering (no gaps):**
- $$\bigcup_{\alpha \in A} T_\alpha = \mathbb{R}^2.$$
2. **Non-overlap:** for every $\alpha \neq \beta$,
- $$\text{int}(T_\alpha) \cap \text{int}(T_\beta) = \emptyset,$$
- where $\text{int}(\cdot)$ denotes the interior.
3. **Topological disks:** each tile T_α is *homeomorphic* to the closed unit disk
- $$D^2 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$$

Fundementals of Penrose Tiling

Penrose tilings are **non-periodic tilings** discovered by mathematician and physicist Roger Penrose in the 1970s. Unlike regular periodic tilings (like squares or hexagons), and Penrose tilings:

- **No Translational Symmetry** – no finite patch of the tiling repeats identically elsewhere.
- **Can be constructed using different rules** – for example, matching rules and projection methods (e.g., de Bruijn's pentagrid approach)

Figure 1. Basic Penrose Rhombic Tiles

Tile Type	Angles	Edge Length
Thin Rhombus	36°, 144°	Equal
Thick Rhombus	72°, 108°	Equal



Figure 2. Basic Penrose Rhombic Tiles

Pentagrid Patterns

Each rhomb has **two pairs of parallel edges**, you can chain rhombs edge-to-edge along any one pair and **create a long “ribbon”**.

There are only **five distinct edge directions** in the Penrose rhomb set, so **each ribbons families do not intersect with each other** since they are parallel.

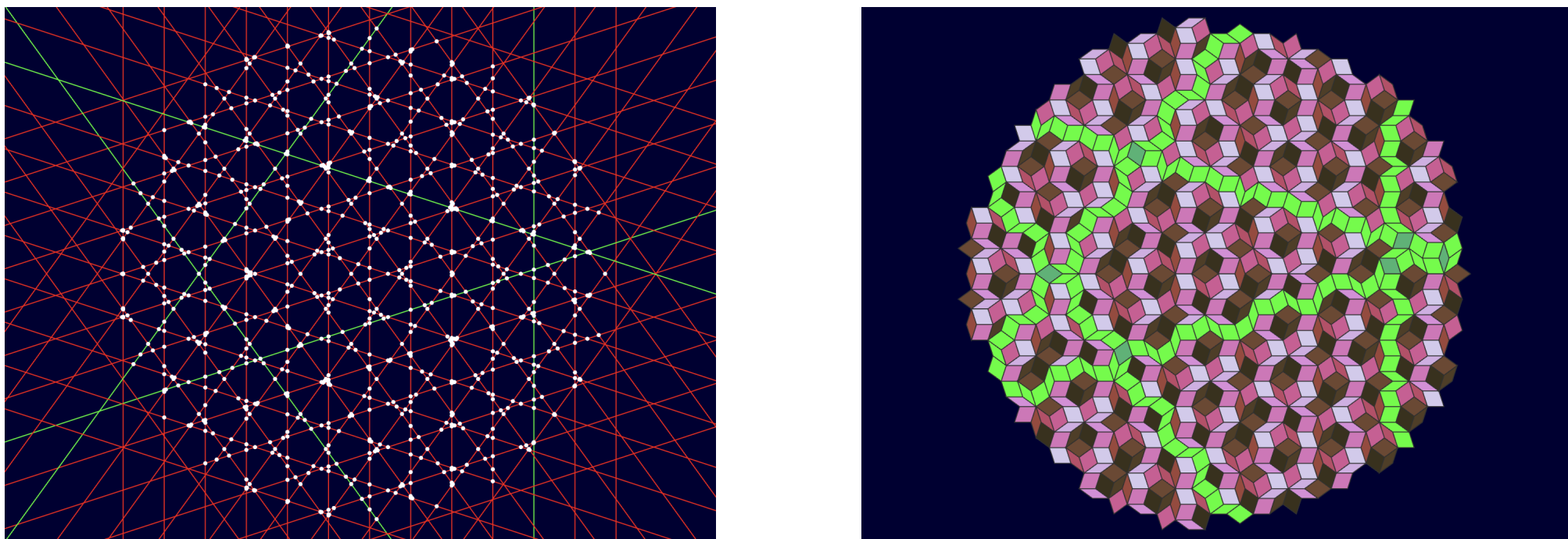


Figure 3. Pentagrid and Tiling

de Bruijn’s found that the ribbons can be reconstructed into sets of parrallel lines, which is called **pentagrid**. While the **intersections of 2 lines** in the pentagrid pattern can be represented a **particular rhomb**.

Constructing Penrose Tilings with Pentagrid

1. **Five grids** For $j = 0, \dots, 4$, set the unit direction vectors

$$v_j = (\cos 2\pi j/5, \sin 2\pi j/5), \quad \gamma_0 + \dots + \gamma_4 = 0.$$

The j -th line family is

$$\mathcal{L}_j = \{x \in \mathbb{R}^2 \mid x \cdot v_j + \gamma_j \in \mathbb{Z}\}.$$

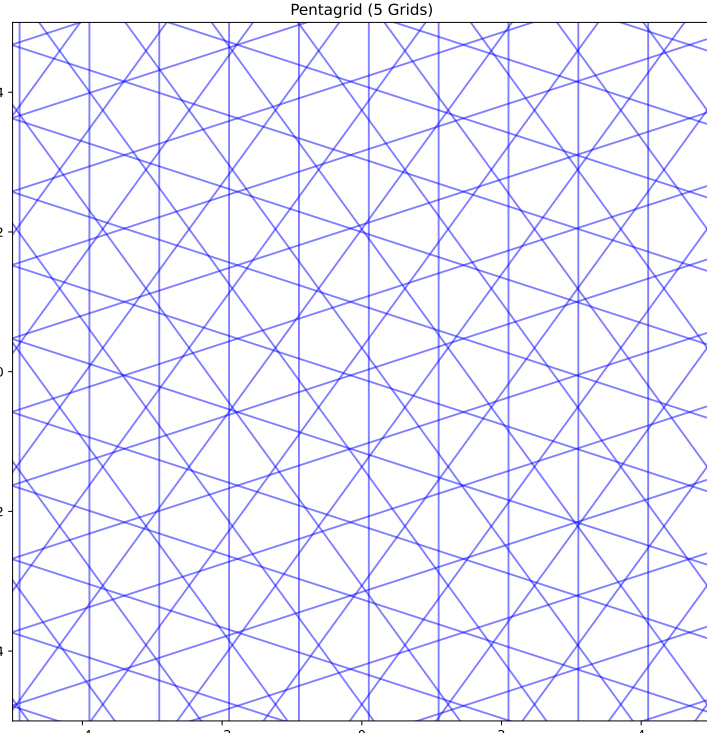
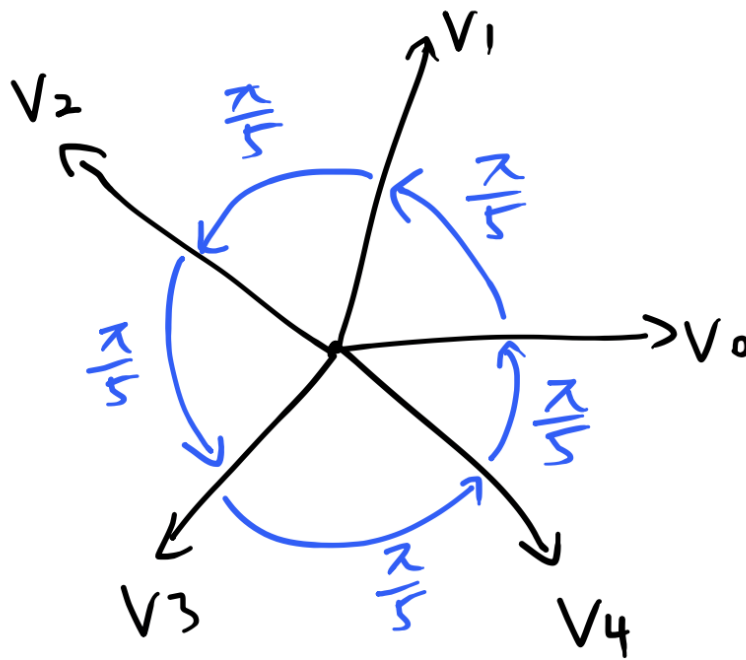


Figure 4. de Bruijn's Pentagrid and its Corresponding Vectors

The pentagrid is called **regular** if no point of C belongs to **more than two of the five grids**, and otherwise singular.

2. **Integer coordinates of a cell** For any $x \in \mathbb{R}^2$, define the integer-valued functions

$$K_j(x) = \lceil x \cdot v_j + \gamma_j \rceil, \quad K(x) = (K_0, \dots, K_4) \in \mathbb{Z}^5.$$

where: γ_j are the grid shift parameters

$K(x) = (K_0(x), \dots, K_4(x))$ is a five-tuple, which are representing the **“pentagrid coordinates of a rhombus in the tiling”**.

The map $x \mapsto K(x)$ is **constant** on each open pentagonal cell and they change only when **crossing a grid line**.

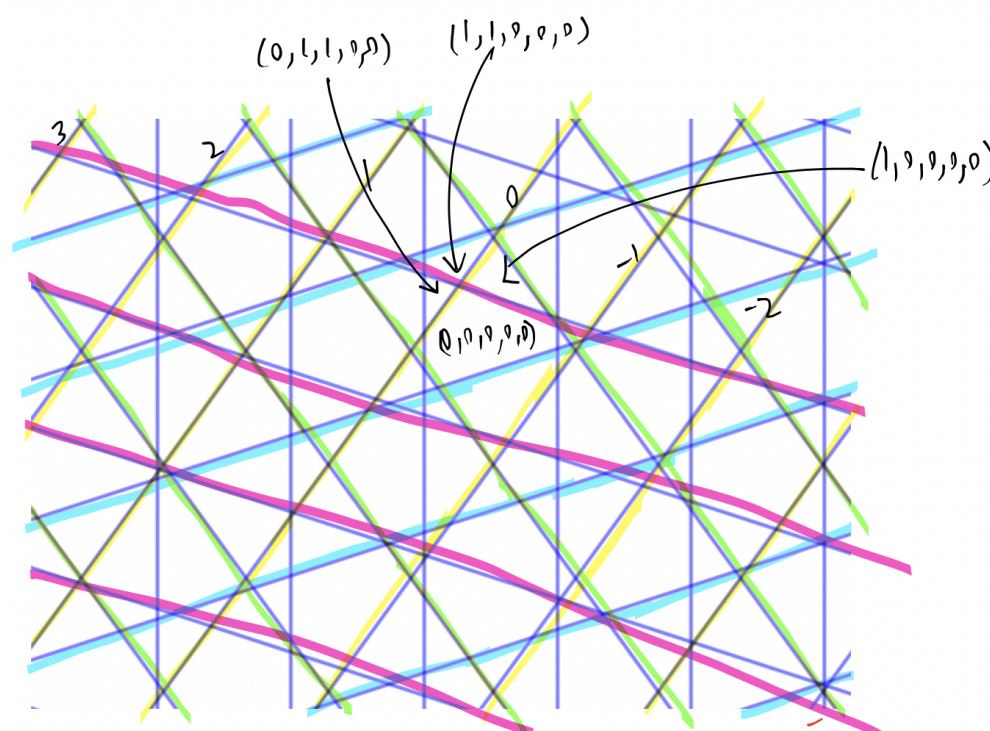
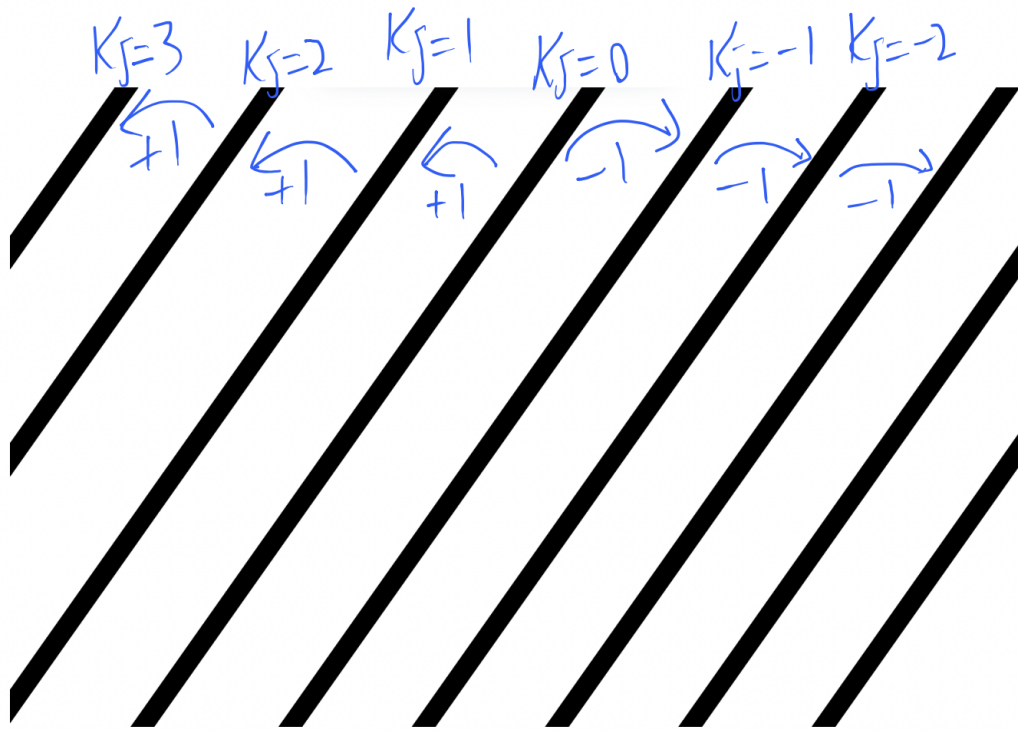


Figure 5. Coordinates in Pentagrid

3. **Linear projection to the plane**

$$f(k_0, \dots, k_4) = \sum_{j=0}^4 k_j v_j$$

The $K(x)$ act as “coordinates” in a 5D space, and $f(x)$ projects them to 2D. The resulting point $f(x)$ is a vertex in the Penrose tiling.

(Continue) Constructing Penrose Tilings with Pentagrid

4. **Each two-line crossing gives one rhomb**. Choose $0 \leq r < s \leq 4$ and integers k_r, k_s . Solving

$$v_r \cdot x + \gamma_r = k_r, \quad v_s \cdot x + \gamma_s = k_s$$

gives an **intersection** x_0 .

For example, given the **five-tuple** $K(x_0) = (K_0, \dots, K_4)$ and the pair $(r, s) = (0, 1)$, the **four vertices** correspond to

$$K(x_0) + (\varepsilon_1, \varepsilon_2, 0, 0, 0), \quad \varepsilon_1, \varepsilon_2 \in \{0, 1\}.$$

Thus, we apply **the linear map** $f(k) = \sum_{j=0}^4 k_j v_j$ to each 5-tuple, which gives

$$f_{\varepsilon_1, \varepsilon_2} = f(K(x_0)) + \varepsilon_1 v_r + \varepsilon_2 v_s, \quad (r, s) = (0, 1).$$

Hence, we have $f_{1,0} - f_{0,0} = v_r$, $f_{0,1} - f_{0,0} = v_s$, so the four vertices (images of the 5-tuple coordinates)

$$\{f_{0,0}, f_{1,0}, f_{1,1}, f_{0,1}\}$$

can **form a parallelogram with edge vectors** $\{v_r, v_s\}$.

Therefore, we can determine whether it is a **thin rhomb** or a **thick rhomb** by computing the angle between v_r and v_s

If angle is 36° or 144°, then the vertices form a **thin rhomb**

If angle is 72° or 108°, then they form a **thick rhomb**

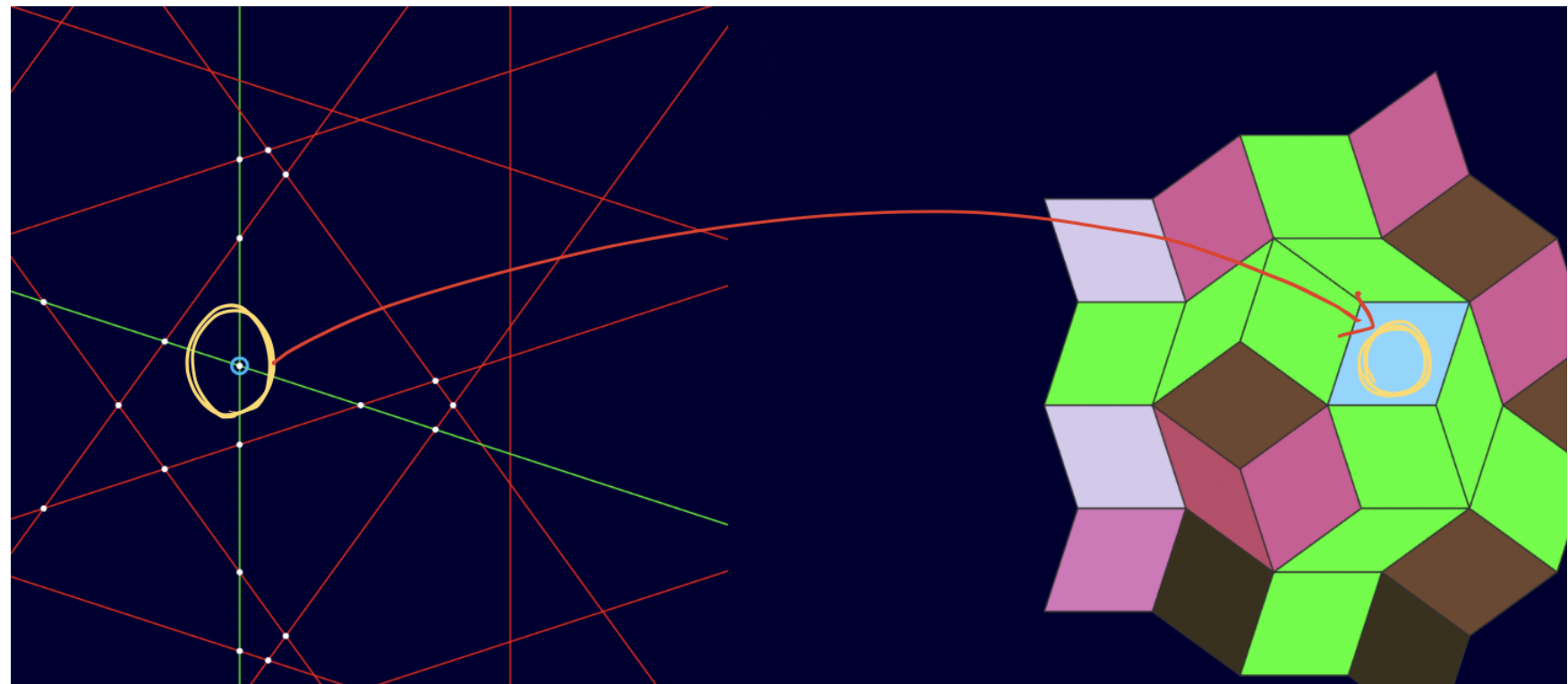
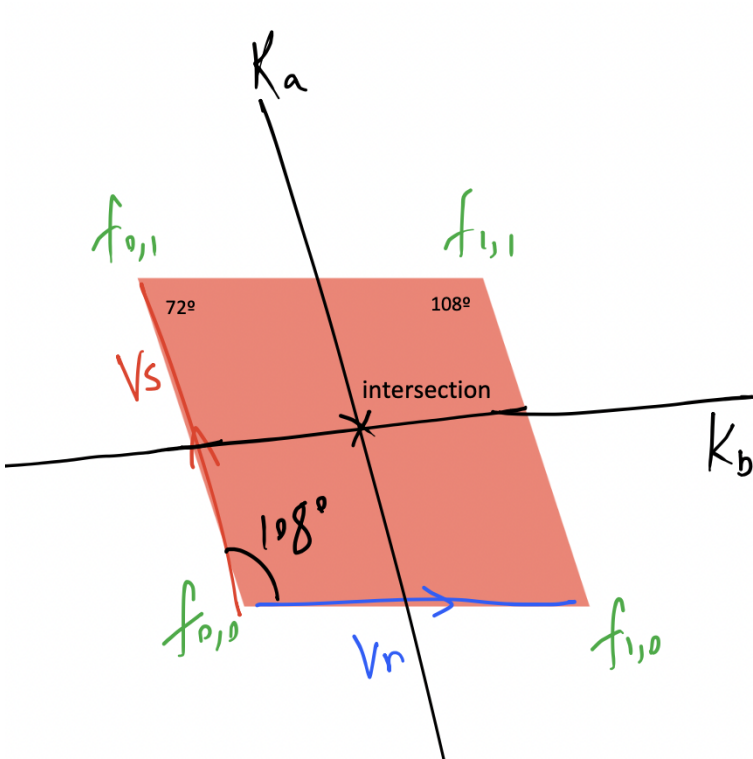


Figure 6. Intersection of two lines with its corresponding rhomb

cell $K(x)$ $\longleftrightarrow$ tiling vertex $f(K)$
intersection of two grids $\longleftrightarrow$ single rhomb

Conclusion

Penrose Tilings can be constructed with different methods and Pentagrid approach is discovered by de Bruijn. The illustrations on the poster are not the only one examples. In fact, there are actually infinitely (**indeed uncountably**) many distinct Penrose tilings, because **sliding any of the five grid families by a real offset** changes the arrangement of tiles but still producing a "legal" Penrose tiling.

References

[1]

T. Treiberg, Penrose Tilings and Quasicrystals (lecture slides). University of Utah, 2023. PDF, www.math.utah.edu/~treiberg/PenroseSlides.pdf.

[2]

E. Mowry B. Shukla, Pentagrids and Penrose Tilings (Hudson River Undergraduate Math Conf.). Williams College, 2016. PDF.

[3]

D. Bailey, Penrose Tilings: A Brief Introduction (course notes). Williams College CS Dept., 2006. PDF.

[4]

N. G. de Bruijn, "Algebraic theory of Penrose's non-periodic tilings of the plane I, II", Indagationes Math. 43 (1981) 39–66 112–128.

[5]

A. Bhatia, "Pattern Collider" (interactive web app), <https://aatishb.com/patterncollider/> (accessed 11 Jun 2025).